

SOOHYUN CHOI

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
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RESEARCH INTERESTS

Reinforcement learning and sequential decision making, with emphasis on iterative policy improvement in offline RL, offline goal-conditioned RL, and generative long-horizon state-space planning.

EDUCATION

Hanyang University <i>Integrated M.S./Ph.D. Program in Electronic Engineering</i> Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong	2024-Present
Hanyang University <i>B.S. in Electronic Engineering</i>	2018-2024

MANUSCRIPT & SELECTED RESEARCH

S. Choi, S. Cho, and S. Hong, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning." *Manuscript, 2026*.

- Developed explicit state-space bridges, inverse-dynamics action chunks, and execution-time replanning for long-horizon offline GCRL; evaluated on OGBench navigation and manipulation tasks.

S. Choi, M. Oh, and I. Song, "ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold." *Undergraduate Capstone Technical Report, 2024; revised 2026*. [\[PDF\]](#)

- Formulated designer-selected performance equivalence classes for circuit search and combined the induced quotient geometry with attention-weighted line search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Multi-step Proximal Policy Improvement • Study offline policy improvement as a short sequence of critic-guided proximal updates whose anchor is re-centered at each step; analyze how the effective step size balances conservative local movement against useful but more critic-error-sensitive finite-step refinement.	2026-Present
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PathFlower - Goal-Conditioned State Flows • Extend PathBridger from an explicit subgoal and bridge toward final-goal-conditioned state flows that generate useful intermediate states for long-horizon planning.	Current
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REACT: Recursive Algorithm for RC Network Transfer Functions <i>First-author industry-sponsored research · SK hynix-Hanyang University project · Unpublished manuscript</i> • Developed a recursive closed-form transfer-function method for arbitrary RC trees and a dominant-pole reduced-order approximation validated against SPICE.	2024-2025
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HONORS & FUNDING

- Top Excellence Award (Hanyang Electronics and Information Communications Alumni Association Special Award) - KRW 1,000,000, 2023
- Top Excellence Award, 15th Capstone Design Fair - KRW 1,000,000, 2023
- Graduation Honors, Hanyang University - 2024
- BK Research Assistantship (BK-RA), Hanyang University - 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I